

Kuiyi (Cory) Gao

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Mandarin (native) | English — TOEFL iBT 110 | Cantonese (intermediate)

Education

University of North Carolina at Chapel Hill

Jan 2026 – present

Undergraduate; intended B.A. in Computer Science

Expected May 2027

- GPA: 3.68/4.00. Exchange student in Spring 2026; transferred in August 2026.
- Coursework: COMP 550 Algorithms and Analysis, COMP 562 Machine Learning, COMP 586 Natural Language Processing, COMP 380 Technology, Ethics & Culture.

The Chinese University of Hong Kong

Sep 2023 – May 2026

Undergraduate study in Computer Science; transferred to UNC

- Faculty of Engineering ELITE Stream. Coursework: ELEG 5491 Introduction to Deep Learning (graduate level), ESTR 3108 Artificial Intelligence, Data Structures, Formal Languages & Automata Theory, Computer Organization & Design, Probability & Statistics, Linear Algebra.

Publications and Manuscripts

[1] Wenxuan Wang, Xiaoyuan Liu, **Kuiyi Gao**, Jen-tse Huang, Youliang Yuan, Pinjia He, Shuai Wang, Zhaopeng Tu. *Can't See the Forest for the Trees: Benchmarking Multimodal Safety Awareness for Multimodal LLMs*. ACL 2025 (Main Conference). [\[paper\]](#) [\[code\]](#)

[2] Wenxuan Wang, **Kuiyi Gao**, Youliang Yuan, Jen-tse Huang, Qiuzhi Liu, Shuai Wang, Wenxiang Jiao, Zhaopeng Tu. *Chain-of-Jailbreak Attack for Image Generation Models via Step by Step Editing*. Findings of ACL 2025. [\[paper\]](#) [\[code\]](#)

[3] Co-first-authored manuscript on *GUI-agent robustness under interface noise*. Under review, 2026.

Research Experience

GUI-Agent Robustness under Interface Noise [3]

Hong Kong & remote

CUHK; advisor: Prof. Michael R. Lyu

Sep 2025 – present

- **Co-first author.** Co-developed a benchmark for web and Android tasks with interface changes such as overlays, moving content, and altered layouts; compared task completion, safety, and action trajectories.
- Designed measures of success under bounded retries, relative degradation, unsafe actions, action differences, and correct action prefixes; contributed to the harness for paired clean and perturbed tasks.

Runtime Safety for Third-Party Agent Skills [\[code\]](#)

Abu Dhabi & Beijing

MBZUAI UGRIP and continuing research; advisor: Dr. Steve Liu

Jun 2026 – present

- Built the runtime monitoring layer in a four-student UGRIP team: compared declared skill behavior with monitored activity and produced PASS, DEFER, or BLOCK verdicts with supporting trace records.
- Developing typed Tickets that bind independently approved permissions, runtime observations, and decisions so an auditor can recompute a verdict. Manuscript in preparation.

Research Experience (continued)

Chain-of-Jailbreak: Image Generation Safety [2]

Hong Kong

Tencent AI Lab & CUHK

Jun 2024 – Feb 2025

- **Second author.** Designed the study, contributed benchmark data, and ran most of the safety and defense experiments on multi-turn image editing.
- Built evaluation pipelines for model responses and a prompting-based defense; contributed to the manuscript and reviewer responses.

MMSafeAware: Multimodal Safety Awareness [1]

Hong Kong & Shenzhen

Tencent AI Lab

Dec 2023 – Oct 2024

- **Third author.** Contributed evaluation design and benchmark construction to a study of 1,500 image-prompt pairs across 29 safety scenarios, covering unsafe and benign inputs.
- Ran vision-centric reasoning fine-tuning experiments within a study of nine multimodal models and three mitigation approaches; evaluated unsafe-input misses and false alarms on benign inputs separately.

Selected Projects

MemSlot: Saliency for Long-Document Compression [code]

2026

UNC COMP 586, Natural Language Processing

- Implemented memory-slot attention over frozen RoBERTa features in PyTorch, with reconstruction and slot-diversity objectives; combined overlapping token windows into word-level saliency.
- Built saliency-based document rendering and an OCR-to-text QA pipeline for CUAD contracts, with text-pruning and summarization controls. End-to-end evaluation is ongoing.

Poetry and Image Variation

2025

CUHK ENGE3940, AI and Language

- Compared CLIP-based variation scores for images generated from ten Imagist and Romantic poetry excerpts, with 20 images per excerpt from each of FLUX.1-dev and Stable Diffusion.
- Used Python to plot score distributions, compare poem groups, and examine the relationship between prompt length and image variation.

Text-KEN: Kernel Comparisons for Sentence Embeddings

May – Aug 2025

CUHK summer research; advisor: Prof. Farzan Farnia

- Built a numerical prototype in Python with MiniLM embeddings, RBF kernel matrices, and spectral entropy; investigated centering and covariance whitening for text novelty analysis.

Technical Skills

Programming and data Python, NumPy, pandas, SciPy; Java and C (coursework).

ML and NLP PyTorch, Hugging Face Transformers, sentence-transformers; attention modules, frozen encoders, and fine-tuning experiments.

Evaluation Benchmark construction; task and trajectory metrics; embedding comparisons; span-QA exact match/F1.

Visualization Matplotlib, seaborn, Pillow; statistical plots, document rendering, and image grids.

Research tools Docker, Playwright, Git, HTTP/JSON APIs; experiment configuration and trace logging.

Honors and Awards

Professor Joseph Lau Scholarship of Innovation; CUHK College Entrance Scholarship; CUHK exchange scholarship (UNC, Spring 2026).